CHAPTER B

# Fluency bank

Two questions per session, in 6 to 8 minutes including the correction. The identifiers allow revisits; they are not a list to be done in full at every lesson. The availability dates are based on the planned teaching, and should be shifted if the topic has not yet been studied.

### R001 – Fractions

Calculate $\frac{2}{3} + \frac{1}{4}$.

### R002 – Order of operations

Calculate $5 - 2(3 - 7)$.

### R003 – Powers

Simplify $\dfrac{10^5 \times 10^{-3}}{10^4}$.

### R004 – Roots

Simplify $\sqrt{50}$.

### R005 – Proportion

18 represents 30 % of a total. What is the total?

### R006 – Direct rate

A price of 80 falls by 15 %. Final value?

### R007 – Initial value

After a 20 % increase, the value is 72. Initial value?

### R008 – Calculated rate

A quantity goes from 40 to 50. Rate?

### R009 – Successive rates

A 20 % increase then a 20 % decrease: overall rate?

### R010 – Reciprocal rate

Which rate compensates a 20 % decrease?

### R011 – Zero product

Solve $(2x-1)(x+3)=0$.

### R012 – Expansion

Expand $(2x-3)^2$.

### R013 – Factorisation

Factorise $3x^2-12x$.

### R014 – Sign of an affine function

Solve $-3x+6\geq 0$.

### R015 – Slope

Slope of the line through $(1;4)$ and $(3;0)$?

### R016 – Line

Equation of the line of slope -2 through $(1;4)$?

### R017 – Weighted mean

The values 2 and 5 have counts 3 and 2. Mean?

### R018 – Median

Median of 1, 2, 2, 5, 8, 10?

### R019 – Quartiles

For the data set 1, 2, 3, 4, 5, 6, 7, 8, give $Q_1$ and $Q_3$ with the ceiling ranks $\left\lceil \frac{n}{4} \right\rceil$ and $\left\lceil \frac{3n}{4} \right\rceil$.

### R020 – Standard deviation of data

For 1 and 3 with equal weights, mean, variance and standard deviation?

### R021 – Units

Convert 1 min 40 s to seconds; 2400 frames over this duration give how many per second?

### R022 – Simple probability

With a fair die, probability of a multiple of 3?

### R023 – Complement

$P(A) = \dfrac{3}{8}$. Give $P(\overline{A})$.

### R024 – Interval

Write $x \geq -2$ and $x < 3$ as an interval.

### R025 – Logic

Negate “all the students passed.

### R026 – Histogram

Two classes of widths 2 and 4 have the same count 20. In a histogram with areas proportional to counts, what are the relative heights?

### R027 – Pie chart

A sector measures 72°. Proportion?

### R028 – Scale

An axis starts at 90. Bars stop at 100 and 110. Is the second value double the first?

### R029 – Percentage of a percentage

40 % of the students take A; among them, 25 % take B. What proportion takes A and B?

### R030 – Proportion and count

A category represents 0.125 of 80 people. Count?

### R031 – Defining and representing a sequence

Calculate the first four terms of $w_0 = 2$, $w_{n+1} = w_n + 2n + 1$; suggest an explicit formula.

### R032 – Arithmetic sequences

We know $u_3 = 17$ and $u_8 = 37$ for an arithmetic sequence. Determine $r$ and $u_0$.

### R033 – Geometric sequences

Study the variation of $u_n = -2 \times 3^n$, then of $v_n = 2(-1)^n$.

### R034 – Sums, indices and proofs

Calculate $\sum_{k=2}^{5} 2^k$ without the summation symbol in the answer.

### R035 – Conjecturing a limit and programming a threshold

For $u_n = 48(\frac{1}{2})^n$, determine the first indices where $u_n \leq 6$ and where $u_n < 6$.

### R036 – Three forms, three readings

Give three forms of $g(x) = x^2 + 2x - 3$. Read off its roots and its minimum.

### R037 – Discriminant and solving

Solve $3x^2 - 6x + 3 = 0$ and $x^2 + 2x + 5 = 0$.

### R038 – Factorising without automating the discriminant

Determine the polynomial f of degree 2 such that $f(1) = f(4) = 0$ and $f(0) = -8$.

### R039 – Sign, inequalities and optimisation

Solve $x^2 - 4x - 5 < 0$, then maximise $x(8 - x)$ on $[0; 3]$.

### R040 – Projection and angle

Two vectors have norms 2 and 5 and form 90°, then 180°. Calculate the products.

### R041 – Coordinates, norm and orthogonality

For which real number t are the vectors $(t; 2)$ and $(3; -6)$ orthogonal?

### R042 – Vector calculus and identities

Given $\|\vec{u}\| = 2$, $\|\vec{v}\| = 3$, $\vec{u} \cdot \vec{v} = 1$. Calculate $\|\vec{u} - \vec{v}\|$.

### R043 – Al-Kashi and choosing a method

A triangle has $AB = AC = 4$ and $BC = 4\sqrt{2}$. Determine the angle A.

### R044 – The circle with diameter AB

For $A(0; 0)$, $B(6; 0)$, determine the points M such that $\overrightarrow{MA} \cdot \overrightarrow{MB} = 7$.

### R045 – Rate of change and secants

Calculate the rate of $f(x) = x^2 + 3x$ between 2 and $2 + h$.

### R046 – Derivative at a point

For $f(x) = 3x^2 - 2x$, calculate $f\prime(1)$ from the rate.

### R047 – Tangent and affine approximation

Given $f(3) = 5$ and $f\prime(3) = -2$. Write the tangent and approximate $f(3.04)$.

### R048 – When differentiability fails

Justify that $g(x) = |x - 2|$ is not differentiable at 2. Also study the differentiability of $x \mapsto \sqrt{x}$ at 0 through its rate.

### R049 – Conditioning: changing the denominator

Out of 60 objects, 20 are red; among the red ones, 5 are damaged. Calculate $P(R \cap D)$ and $P_R(D)$.

### R050 – Trees, partitions and total probability

$P(A) = \dfrac{1}{3}$, $P_A(B) = \dfrac{3}{4}$, $P_{\overline{A}}(B) = \dfrac{1}{4}$. Calculate $P(B)$.

### R051 – Independence and disjointness

Given $P(A) = 0.4$, $P(B) = 0.5$, $P(A \cap B) = 0.18$. Are they independent?

### R052 – Successive trials and Bernoulli trials

Four independent trials, each with probability $p = \dfrac{1}{3}$. Calculate the probability of four failures, then of at least one success.

### R053 – Simulating and discussing the model

A simulation of 2000 points gives 675 points below $y = x^2$. What area does it estimate?

### R054 – Derivative function and standard functions

Differentiate $x^5$, $x^{-2}$ and $\sqrt{x}$, stating the domains.

### R055 – Sum, product, reciprocal and quotient

Differentiate $(x^2 + 1)(3x - 2)$, then $\dfrac{1}{x^2 + 1}$.

### R056 – Sign of the derivative and extrema

Study $f(x) = x^2 - 4x + 1$ on $[0; 3]$ and give its extrema on this interval.

### R057 – Parity, inequalities and comparison

Study the parity of $x^4 + x^2$ and $x^3 - x$; prove $x^2 + 1 \geq 2x$.

### R058 – Optimisation and Newton's method

A rectangle of perimeter 12 m has width x. Determine the maximum area by differentiation.

### R059 – Normal vector and Cartesian equation

Equation of the line through $(2; -1)$ with normal $(3; 4)$.

### R060 – Orthogonal projection and distance

Project $P(4; 1)$ onto $d : x = 2$.

### R061 – Equation of a circle

Recognise $x^2 + y^2 + 2x - 8y + 17 = 0$.

### R062 – Solving a configuration

Intersect $(x - 1)^2 + (y + 2)^2 = 9$ with $y = 1$.

### R063 – Variable, event and distribution

Two fair coins are tossed. X counts the heads. Write its distribution and $P(X \leq 1)$.

### R064 – Expectation and linearity

A gross gain is 10 with probability $\dfrac{1}{5}$ and 0 otherwise. What stake makes the game fair?

### R065 – Variance, standard deviation and König–Huygens

For X with values $-1, 1$ equally likely, calculate the expectation, variance and standard deviation.

### R066 – Transforming and comparing gains

$E(X) = 3$, $V(X) = 4$. Give the characteristics of $Y = -2X + 5$.

### R067 – Samples and observed mean

For $\mu = 5$, $\sigma = 3$ and n=36, give the experimental interval asked for.

### R068 – Defining a new function

What is the equation of the tangent to $e^x$ at 1?

### R069 – Algebraic properties and comparisons

Simplify $\dfrac{e^{3x}}{e^{x-1}}$; solve $e^{x+2} > e^{2x}$.

### R070 – Differentiating and studying

Differentiate $g(t) = 3e^{-2t}$ and study its direction of variation.

### R071 – From discrete to continuous

For $Q(t) = 5e^{0.3t}$, express $Q(n)$ and $\dfrac{Q(t+2)}{Q(t)}$.

### R072 – Building numerically and proving an inequality

Show that $e^{2x} \geq 1 + 2x$ for every real number x.

### R073 – Measuring an angle in radians

Convert 45°, 120°; give the arc for radius 2 m and angle $\dfrac{3\pi}{4}$.

### R074 – Wrapping and the image of a real number

Place the images of $\dfrac{7\pi}{2}$ and $-\dfrac{5\pi}{3}$ by reducing them to within one turn.

### R075 – Sine, cosine and standard values

If $\cos x = \dfrac{3}{5}$ and the image of x is in the fourth quadrant, give $\sin x$.

### R076 – Related angles and symmetries

Calculate the two coordinates of the images of $\dfrac{4\pi}{3}$ and $\dfrac{7\pi}{4}$.

### R077 – Approximating pi and putting it to use

For n=6 and c=1, calculate the new chord after doubling.

## R078 – Reading a graph

On the figure R078, read $f(1)$, solve $f(x) = 0$ and $f(x) > 0$.

